Pd(0)-Catalyzed Insertion−Cyclization Reaction of 2,3-Allenols with Aryl or Alkenyl Halides. Diastereoselective Synthesis of Highly Optically Active *trans*-2,3-Disubstituted Vinylic Oxiranes

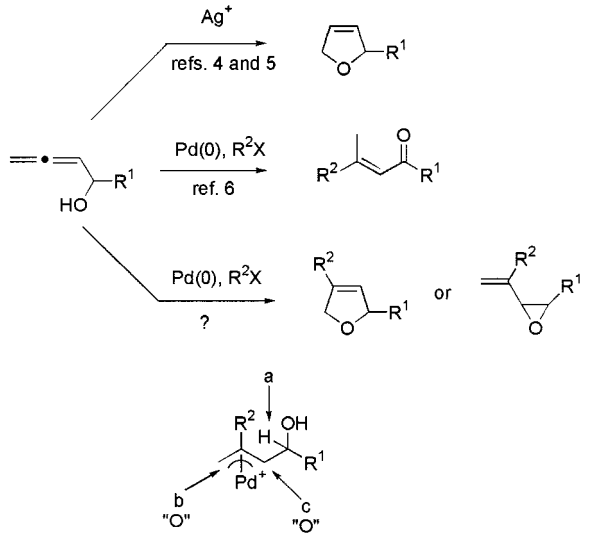
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Allenes are a class of compounds with unique reactivities due to the existence of the two orthogonal π-bonds, and they have been found to be very useful intermediates in organic synthesis.[1−3] Among our efforts in the area of allene chemistry,[3] we developed an efficient one-step route to butenolides with potential biological activities via the Pd(0)-catalyzed cyclization reaction of 1,2-allenic carboxylic acids with aryl or 1-alkenyl halides.[3c] In this paper we give our results on the corresponding reaction of 2,3-allenols (Scheme 1).

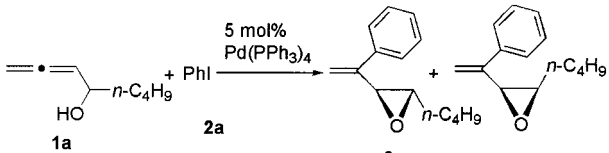
Scheme 1



The groups of Olsson[4] and Marshall[5] have shown that the Ag^+-catalyzed cyclization reaction of 2,3-allenols affords 2,5-dihydrofuran derivatives, while Tsuji and co-workers[6] reported that the Pd(0)-catalyzed reaction of 2,3-dienols with aryl or alkenyl halides produced β-aryl- or alkenyl-β-methyl-α,β-unsaturated enones via an insertion−β-elimination mechanism (pathway a, Scheme 1). We are interested in the possibility of the Pd(0)-catalyzed reaction of organohalides and 2,3-allenols to form either 2,4-disubstituted-2,5-dihydrofurans (pathway b, Scheme 1) or 2,3-disubstituted vinylic oxiranes (pathway c, Scheme 1).

In our first attempt, we synthesized 1,2-octadien-4-ol[7] and studied its Pd(0)-catalyzed cyclization reaction with PhI under various reaction conditions (Scheme 2). Luckily and surprisingly, we observed in our first try the formation of the *three-membered ring*, i.e., 2-(1′-phenylethenyl)-3-(*n*-butyl)oxirane **3a**, under the conditions for the corresponding cyclization reaction of 1,2-allenyl carboxylic acids (Scheme 2).[3c] The formation of the 2,5-dihydrofuran derivative was not observed, and the Ag^+ salt was not required (compare entries 1 with 2 and 3 with 4, Scheme 2). After we screened several different reaction conditions, the stereoselectivity of formation of the two diastereoisomers was increased to 30:1 by using DMF as the solvent (compare entries 1−4 with 5, Scheme 2). The relative configuration of the major isomer was established to be *trans*, as determined from the NOESY spectra of **3e** (entry 5, Table 1).

Scheme 2[a]



entry	conditions	Yield (%)[a] of **3a** +**3a'**	**3a**/**3a'** [a]
1	K_2CO_3, Ag_2CO_3, CH_3CN, reflux, 18 h	67[b]	18:1
2	K_2CO_3,CH_3CN, reflux, 18 h	62	14:1
3	K_2CO_3, Ag_2CO_3, CH_3CN, 65 °C, 18 h	33[c]	15:1
4	K_2CO_3, CH_3CN, 65°C, 18 h	32[d]	20:1
5[e]	K_2CO_3, DMF, 55 °C, 14 h	71[b]	30:1

[a] Determined from 300 MHz ^{1}H NMR spectra of the crude product before separation using CH_2Br_2 as the internal standard. [b] Isolated yield. [c] 23% of **1a** was recovered. [d] 18% of **1a** was recovered. [e] This set of reaction conditions is defined as conditions A.

Using conditions A (entry 5, Scheme 2) as the standard reaction conditions, we studied the Pd(0)-catalyzed highly diastereoselective epoxide formation reaction carefully, and some of our typical examples were listed in Table 1. The following points should be mentioned:

(1) In most cases, the diastereoselectivities of this reaction are fairly high.

(2) Both aryl and 1-alkenyl halides could afford the corresponding products in moderate to good yields (for the reaction with 1-alkenyl iodides, see entries 4 and 11, Table 1).

(3) For the reaction of *p*-bromoiodobenzene, only the carbon−iodine bond reacted, while the carbon−bromine bond remained intact (entries 10 and 15, Table 1).

Optically active epoxides are considered as important synthetic intermediates.[8] Jung et al. studied the rearrangement of vinylic

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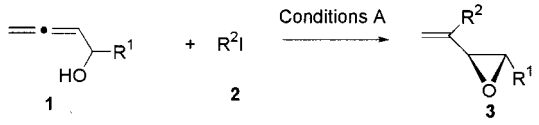
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Table 1. Pd(0)-Catalyzed Insertion−Cyclization Reaction of 2,3-Allenol and Aryl or Alkenyl Halides



1 2 3

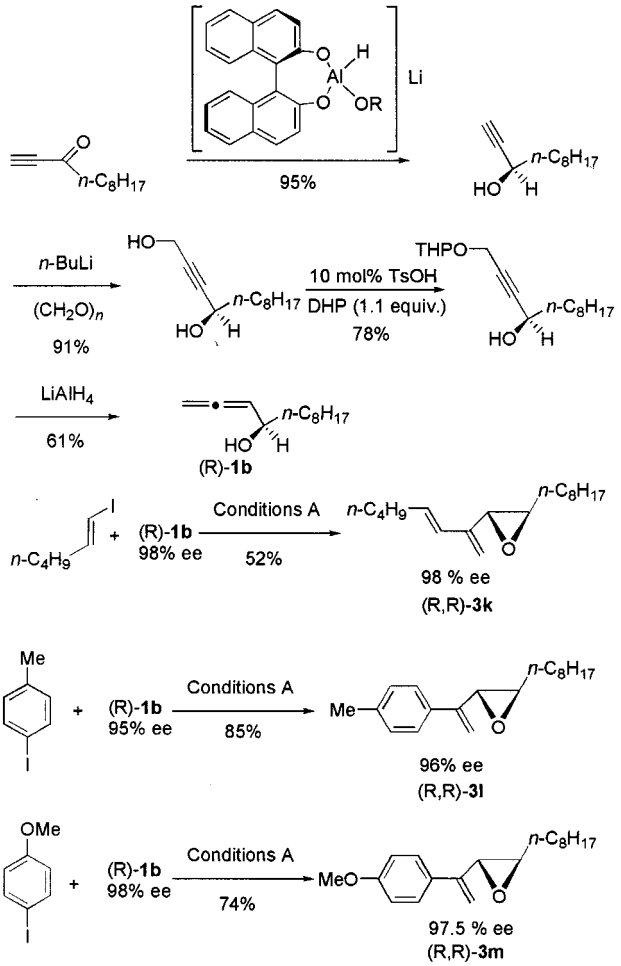
entry	**1**, R^1	**2**, R^2	temp (°C)	time (h)	product	yield (%)	dr[a]
1	*n*-C_4H_9	Ph	55	14	**3a**	71	96/4
2	*n*-C_4H_9	*p*-MeO-C_6H_4	52	16	**3b**	71	97/3
3	*n*-C_4H_9	*p*-Me-C_6H_4	53	16	**3c**	70	97/3
4	*n*-C_4H_9	*E*-2-Ph-ethenyl	54	24	**3d**	65	92/8[b]
5	*n*-C_4H_9	naphthyl	64	48	**3e**	76	96/4
6	Ph	*p*-Me-C_6H_4	52	16	**3f**	55	95/5
7	Ph	*p*-MeO-C_6H_4	53	16	**3g**	66	92/8[b]
8	*n*-C_7H_{15}	*p*-Me-C_6H_4	54	16	**3h**	88	97/3
9	*n*-C_7H_{15}	*p*-MeO-C_6H_4	52	14	**3i**	81	99/1
10	*n*-C_7H_{15}	*p*-Br-C_6H_4	56	48	**3j**	80	98/2
11	*n*-C_8H_{17}	*E*-1-hexenyl	54	20	**3k**	59	*c*
12	*n*-C_8H_{17}	*p*-Me-C_6H_4	54	16	**3l**	95	97/3[d]
13	*n*-C_8H_{17}	*p*-MeO-C_6H_4	54	16	**3m**	83	98/2[d]
14	*n*-C_8H_{17}	Ph	58	18	**3n**	78	98/2
15	*n*-C_8H_{17}	p-Br-C_6H_4	56	18	**3o**	46	99/1[e]

[a] Diastereomer ratio; determined from 300 MHz ^{1}H NMR spectra of the crude products before separation. In all cases, except entries 4 and 7, the two isomers could not be separated via chromatography on silica gel using the conditions described in this paper. [b] The minor isomer could be isolated by chromatography on silica gel. [c] *Cis* isomer not observed. [d] The data were also confirmed by HPLC. [e] Determined from 400 MHz ^{1}H NMR spectrum.

epoxides for the synthesis of α-alkyl amino aldehydes and acids.[9] With our protocol for the highly diastereoselective synthesis of *trans*-2,3-disubstituted vinylic epoxides in hand, we utilized the highly enantioselective reduction of 1-undecyn-3-one to make optically active (*R*)-1-undecyn-3-ol,[10] which can be easily converted to the optically active (*R*)-1,2-dodecadien-4-ol (**1b**) (Scheme 3).[7] Its cyclization reaction with (1*E*)-hexenyl iodide, *p*-methylphenyl iodide, and *p*-methoxyphenyl iodide afforded (*R*,*R*)-**3k**, (*R*,*R*)-**3l**, and (*R*,*R*)-**3m**, respectively (Scheme 3).[11]

In conclusion, we have developed a mild, good-yielding, and highly diastereoselective methodology for the synthesis of *trans*-2,3-disubstituted vinylic oxiranes. Using the readily available chiral 2,3-allenols,[12] the corresponding vinylic oxiranes with high ee value can be synthesized.[13−15] Due to the wide synthetic importance of vinylic oxiranes, this methodology will show its utility in organic synthesis.

Scheme 3



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Supporting Information Available: Experimental details and ^{1}H NMR spectra of **3a**−**o** (PDF). This material is available free of charge via the Internet at http://pubs.acs.org.

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